INCH-POUND
MS19068C
14 March 2023
SUPERSEDING
MS19068B
13 February 2020

DETAIL SPECIFICATION SHEET

NUTS, PLAIN, ROUND, RETAINING, BALL AND ROLLER BEARING, REGULAR SERIES

Reactivated after 14 March 2023 and may be used for new and existing designs and acquisitions

This specification is approved for all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-DTL-21337.

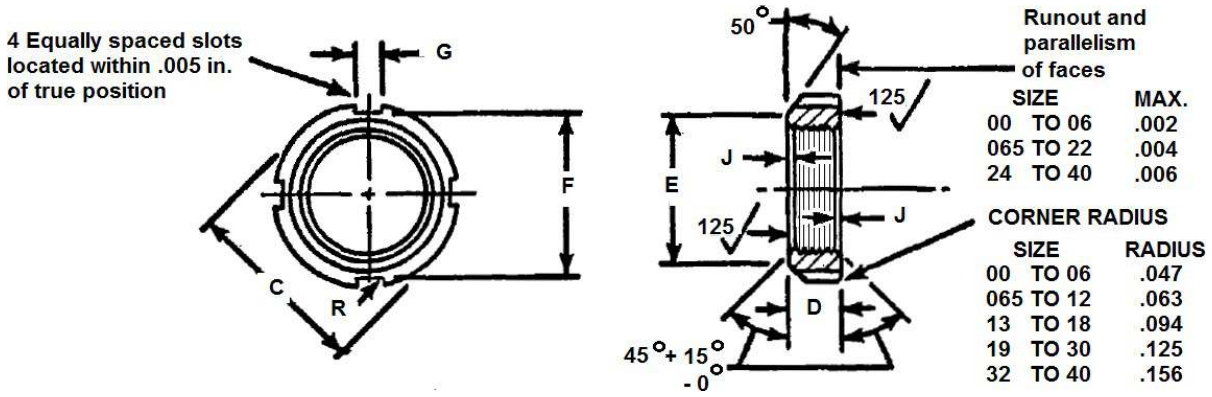


FIGURE 1. Nut.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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TABLE I. Dash numbers and dimensions

Dash No.	D Thickness		C 0D 0.005 -0.015	E Face Dim		J Cham	Slot dimensions F 0.01 -0.02	G		R	Thread data No. per inch	Minor dia		Pitch dia		Major dia
	Min	Max		Min	Max			Min	Max	Max		Min	Max	Min	Max	Min
00	0.209	0.229	0.750	0.605	0.625	0.031	0.625	0.120	0.130	0.010	32	0.3572	0.3606	0.3707	0.3733	0.391
01	0.303	0.323	0.875	0.699	0.719	0.031	0.750	0.120	0.130	0.010	32	0.4352	0.4386	0.4487	0.4513	0.469
02	0.303	0.323	1.000	0.793	0.813	0.031	0.812	0.120	0.130	0.010	32	0.5522	0.5556	0.5657	0.5687	0.586
03	0.334	0.354	1.125	0.918	0.938	0.031	0.937	0.120	0.130	0.010	32	0.6302	0.6336	0.6437	0.6467	0.664
04	0.365	0.385	1.375	1.105	1.125	0.031	1.187	0.178	0.198	0.015	32	0.7472	0.7506	0.7607	0.7641	0.781
05	0.396	0.416	1.562	1.261	1.281	0.031	1.375	0.178	0.198	0.015	32	0.9352	0.9386	0.9487	0.9521	0.969
06	0.396	0.416	1.750	1.480	1.500	0.047	1.562	0.178	0.198	0.015	18	1.1129	1.1189	1.1369	1.1409	1.173
065	0.428	0.448	2.062	1.793	1.813	0.047	1.875	0.178	0.198	0.015	18	1.2524	1.2584	1.2764	1.2804	1.313
07	0.428	0.448	2.062	1.793	1.813	0.047	1.875	0.178	0.198	0.015	18	1.3159	1.3219	1.3399	1.3439	1.376
08	0.428	0.448	2.250	1.980	2.000	0.047	2.062	0.240	0.260	0.020	18	1.5029	1.5089	1.5269	1.5314	1.563
09	0.428	0.448	2.531	2.261	2.281	0.047	2.343	0.240	0.260	0.020	18	1.7069	1.7129	1.7309	1.7354	1.767
10	0.49	0.51	2.687	2.418	2.438	0.047	2.500	0.240	0.260	0.020	18	1.9069	1.9129	1.9309	1.9354	1.967
11	0.49	0.51	2.968	2.636	2.656	0.047	2.718	0.240	0.260	0.020	18	2.0969	2.1029	2.1209	2.1260	2.157
12	0.521	0.541	3.156	2.824	2.844	0.047	2.906	0.240	0.260	0.020	18	2.2999	2.3059	2.3239	2.3290	2.360
13	0.553	0.573	3.375	3.043	3.063	0.047	3.125	0.240	0.260	0.020	18	2.4879	2.4939	2.5119	2.5170	2.548
14	0.553	0.573	3.625	3.283	3.313	0.047	3.375	0.240	0.260	0.020	18	2.6909	2.6969	2.7149	2.7200	2.751
15	0.548	0.0604	3.875	3.533	3.563	0.078	3.625	0.360	0.385	0.025	12	2.8428	2.8518	2.8789	2.8843	2.933
16	0.584	0.604	4.156	3.814	3.844	0.078	3.906	0.360	0.385	0.025	12	3.0468	3.0558	3.0829	3.0888	3.137
17	0.615	0.635	4.406	4.001	4.031	0.078	4.093	0.360	0.385	0.025	12	3.2498	3.2588	3.2859	3.2918	3.340
18	0.678	0.698	4.656	4.251	4.281	0.078	4.343	0.360	0.385	0.025	12	3.4368	3.4458	3.4729	3.4803	3.527
19	0.709	0.729	4.937	4.533	4.563	0.078	4.625	0.360	0.385	0.025	12	3.6398	3.6488	3.6759	3.6833	3.730
20	0.735	0.76	5.187	4.783	4.813	0.078	4.875	0.360	0.385	0.025	12	3.8278	3.8368	3.8639	3.8713	3.918
21	0.735	0.76	5.437	4.970	5.000	0.078	5.062	0.485	0.510	0.030	12	4.0318	4.0408	4.0679	4.0762	4.122
22	0.766	0.791	5.718	5.251	5.281	0.078	5.343	0.485	0.510	0.030	12	4.2348	4.2438	4.2709	4.2792	4.325
24	0.798	0.823	6.125	5.658	5.688	0.078	5.750	0.485	0.510	0.030	12	4.6258	4.6348	4.6619	4.6702	4.716
26	0.860	0.885	6.750	6.158	6.188	0.078	6.250	0.610	0.635	0.030	12	5.0158	5.0248	5.0519	5.0602	5.106
28	0.923	0.948	7.093	6.501	6.531	0.078	6.593	0.610	0.635	0.030	12	5.4068	5.4158	5.4429	5.4512	5.497
30	0.954	0.979	7.687	7.033	7.063	0.078	7.125	0.610	0.635	0.030	12	5.7978	5.8068	5.8339	5.8422	5.888
32	1.016	1.041	8.062	7.398	7.438	0.094	7.500	0.610	0.635	0.030	8	6.1487	6.1622	6.2028	6.2119	6.284
34	1.048	1.073	8.656	7.991	8.031	0.094	8.093	0.610	0.635	0.030	8	6.5237	6.5372	6.5778	6.5869	6.659
36	1.079	1.104	9.062	8.335	8.375	0.094	8.437	0.735	0.760	0.030	8	6.9307	6.9442	6.9848	6.9939	7.066
38	1.110	1.135	9.468	8.741	8.781	0.094	8.843	0.735	0.760	0.030	8	7.3367	7.3502	7.3908	7.3999	7.472
40	1.173	1.198	9.843	9.116	9.156	0.094	9.218	0.735	0.760	0.030	8	7.7117	7.7252	7.7658	7.7772	7.847

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NOTES:

1. MATERIALS:

   (a) Carbon steel, SAE 1015 to 1035 (UNS G10150 to UNS G10350), 1112 to 1118 (UNS G11120 to UNS G11180) or 1212 to 1215 (UNS G12120 to UNS G12150) in accordance with SAE J403.

   (b) Corrosion resisting steel: 302 (UNS S30200), 303 (UNS S30300) or 304 (UNS S30400) in accordance with ASTM A959.

2. HARDNESS:

   (a) Carbon steel nuts shall have a hardness of Rockwell HRB 66 to 92.

   (b) Corrosion resisting steel shall have a maximum hardness of Rockwell HRB 90.

3. PROTECTIVE COATING:

   (a) Carbon steel nuts shall be either:
   - zinc-nickel plated in accordance with MIL-PRF-32647, Type K20
   - zinc plated in accordance with ASTM F1941/F1941M Fe/Zn 5BNS
   - or uncoated.

   (b) Corrosion resisting steel nuts shall be passivated in accordance with ASTM A967/A967M, Nitric treatment 2-5 or Citric treatment 1-5, or SAE AMS2700, Method 1, Type 6, 7 or 8, or Method 2. Hexavalent chromium containing compounds shall not be used in the passivation treatment nor in post-cleaning treatments. Hexavalent chromium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List.

4. TESTING: MIL-STD-1916, Test Method Standard, shall be utilized to test the fasteners for quality and inspection frequency.

5. THREADS: Threads shall be American National Form, Class UNC-3B, in accordance with FED-STD-H28.

6. SURFACE ROUGHNESS: Surface roughness shall be in accordance with ASME B46.1.

7. MAGNETIC PARTICLE INSPECTION: Inspection shall be in accordance with the procurement specification.

8. LIQUID PENETRANT INSPECTION: Inspection for corrosion resisting steel parts shall be in accordance with ASTM E1417/E1417M.

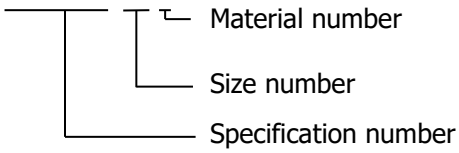
9. DIMENSIONS: All dimensions are in inches. Tolerances; decimals ± 0.010, angles ± 2º unless otherwise specified.

10. PART NUMBER: The MS part number is the MS number, plus the size and the material number. The material numbers are:

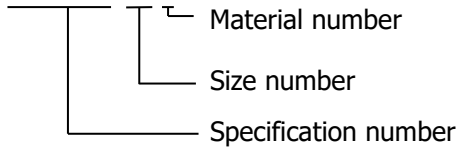
    1 – carbon steel (plain)
    2 – zinc plated carbon steel
    3 – corrosion resisting steel (passivated)
    4 – zinc-nickel plated carbon steel

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Examples: MS19068-00 2



MS19068-05 4



11. WASHER: Use MS19070 for applicable key washers. Applicable washers shall have size numbers corresponding to nut size numbers: Example: nut size 00 should be used with washer size number 00.

12. For design feature purpose, this specification takes precedence over procurement documents referenced herein.

13. Referenced documents shall be of the issue in effect on the date of invitations for bids, or request for proposal, except that referenced adopted industry documents shall give the date of the issue adopted.

14. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy – OS
DLA – GS

Preparing activity:
Army – AT

(Project 5310-2023-010)

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